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Fundamental Concepts and Methods in Biomedical Image Registration

Group 15 – Final Presentation

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ONE LOVE. ONE FUTURE.

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1. Introduction: What is Biomedical Image Registration?

- **Aligning** two medical images so that the **same anatomical structures** correspond to each other in space.

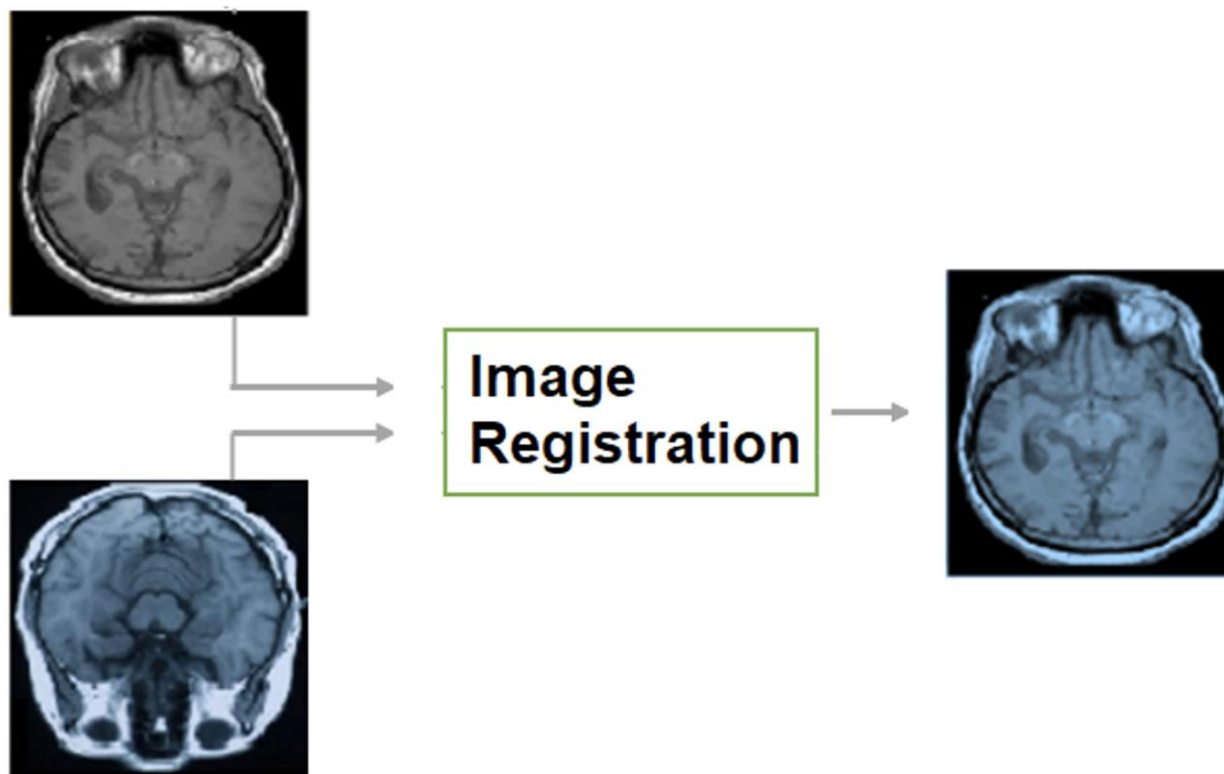


Figure 1. Example illustration of the Biomedical Image Registration Problem¹

1. Introduction: Applications

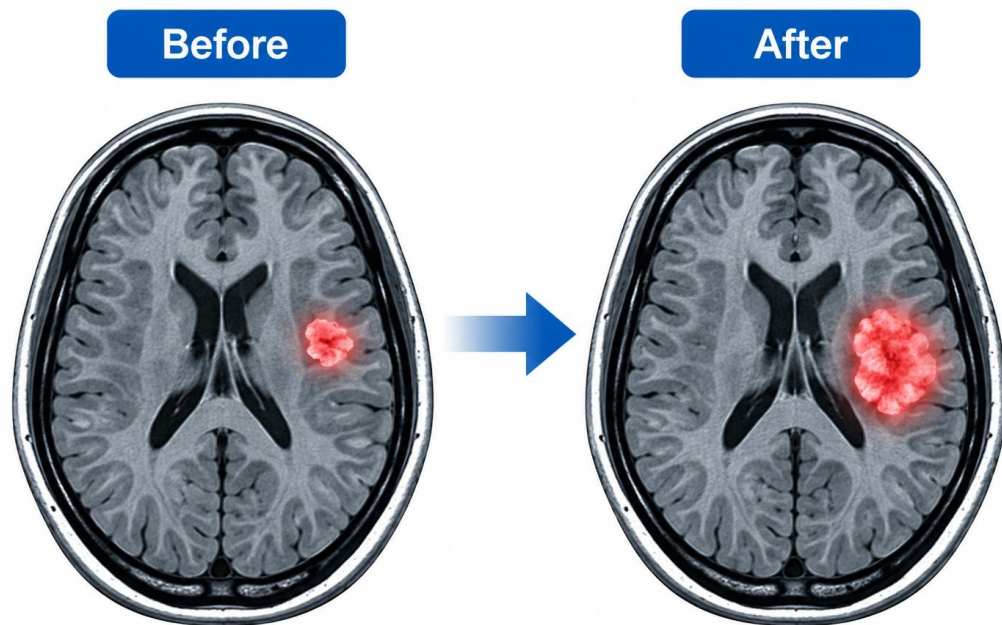


Figure 2. **Disease monitoring:** track tumor changes and disease progression¹

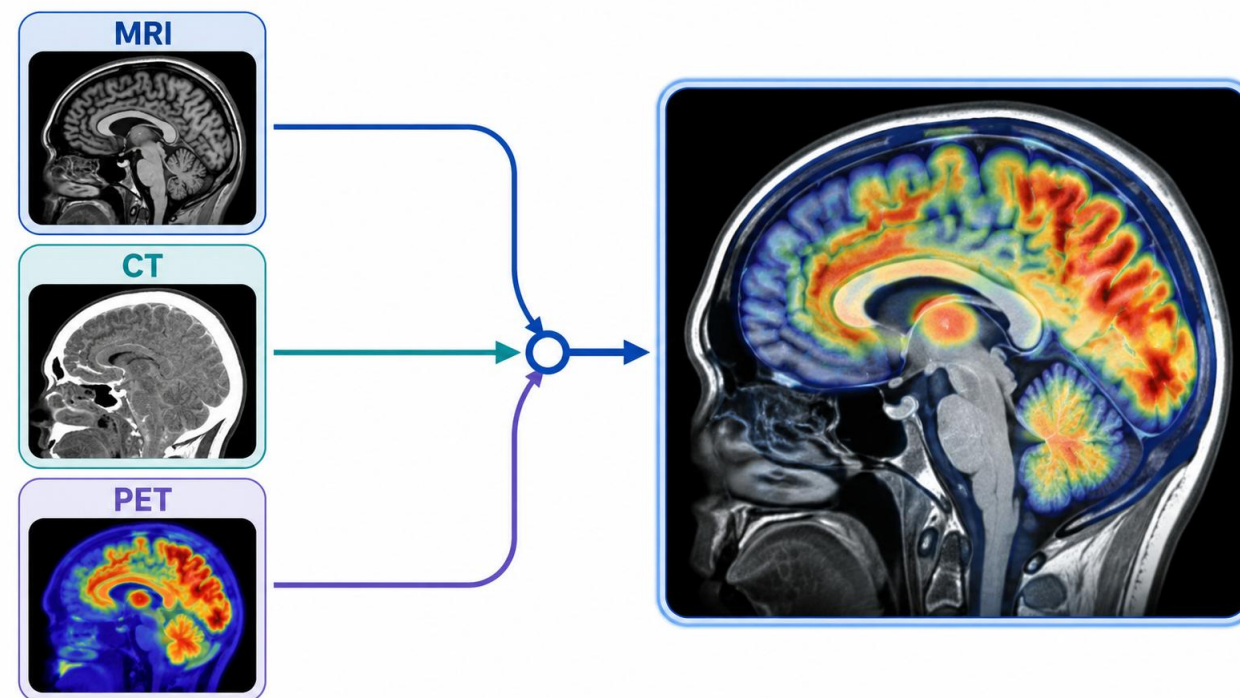


Figure 3. **Multimodal fusion:** combine modalities for richer anatomical and functional information²

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2. Problem Overview: Definitions

- Image domain:

$$\Omega \subset \mathbb{R}^d, d \in \{2, 3\}$$

- Fixed image:

$$F : \Omega \rightarrow \mathbb{R}$$

- Moving image:

$$M : \Omega \rightarrow \mathbb{R}$$

- Spatial transformation:

$$\phi : \Omega \rightarrow \Omega$$

- Transformed moving image:

$$(M \circ \phi)(x) = M(\phi(x))$$

2. Problem Overview: Formulation

- **Objective:** Find the optimal transformation that aligns the moving image with the fixed image.

Optimal transformation:

The best spatial mapping found by registration.

Dissimilarity term:

Measures how different the transformed moving image is from the fixed image.

$$\phi^* = \arg \min_{\phi \in \mathcal{T}} [D(F, M \circ \phi) + \lambda R(\phi)]$$

Transformation space:
All allowed transformations

Regularization term:

Encourages the transformation to be smooth and anatomically plausible.

2. Problem Overview: Transformation Models

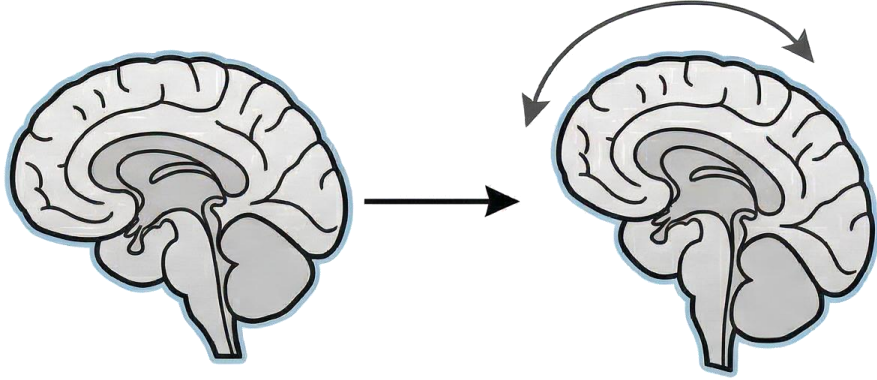


Figure 4. Rigid Transformation¹

$$\phi(x) = Rx + t$$

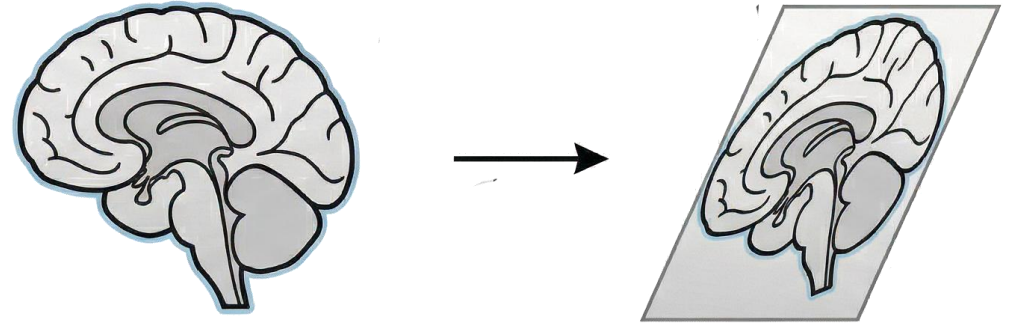


Figure 5. Affine Transformation²

$$\phi(x) = Ax + b$$

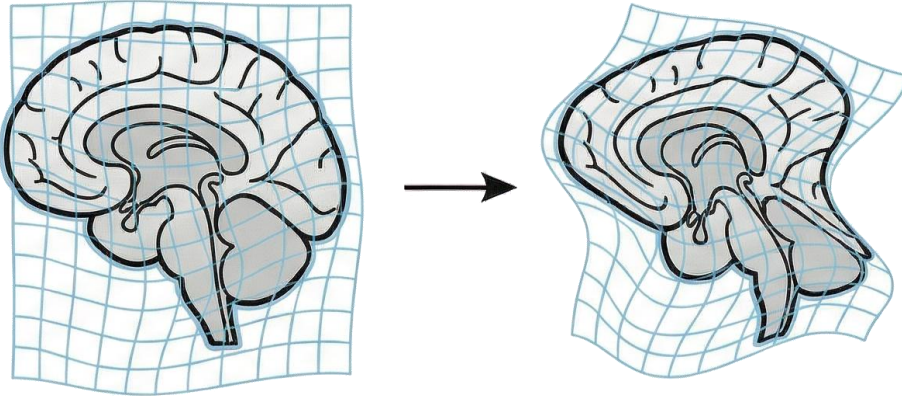


Figure 6. Deformable Transformation³

$$\phi(x) = x + u(x)$$

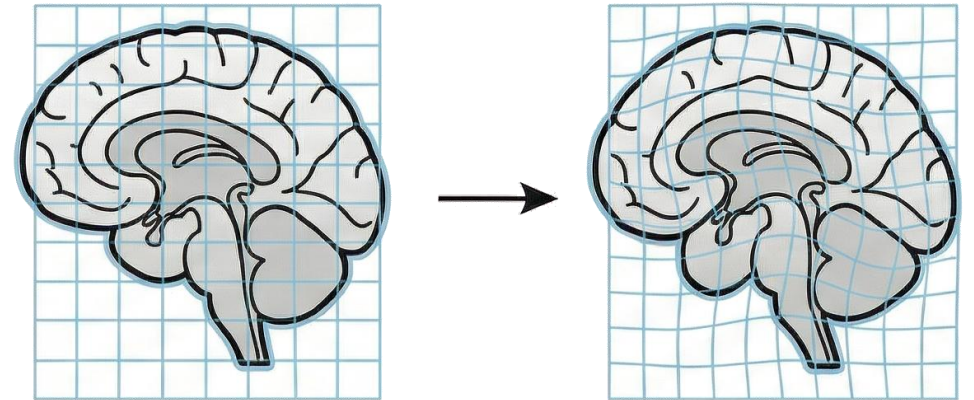


Figure 7. Diffeomorphic Transformation⁴

$$\phi^{-1} \text{ exists}$$

2. Problem Overview: Dissimilarity term

- Measures how well the warped moving image matches the fixed image.

$$D(F, M \circ \phi)$$

- Mean Squared Error:** Directly compares voxel intensity values.

$$D_{\text{MSE}} = \frac{1}{|\Omega|} \sum_{x \in \Omega} [F(x) - M(\phi(x))]^2$$

- Normalized Cross-Correlation:** correlation between intensity patterns.

$$D_{\text{NCC}} = \frac{\left\langle F(x) - \bar{F}, M(\phi(x)) - \overline{M(\phi(x))} \right\rangle_{\Omega}}{\|F(x) - \bar{F}\|_{\Omega} \|M(\phi(x)) - \overline{M(\phi(x))}\|_{\Omega}}$$

2. Problem Overview: Regularization

- Prevents **unrealistic** deformation
- Keeps the transformation anatomically **reasonable**
- λ : trade-off between image matching and regularization

$$\lambda R(\phi)$$

Smoothness:

Neighboring points should move in similar directions

$$R_{\text{smooth}}(u) = \int_{\Omega} \|\nabla u(x)\|^2 dx$$

Bending energy:

Penalizes sharp bending in the deformation field

$$R_{\text{bend}}(u) = \int_{\Omega} \sum_{i,j} \left(\frac{\partial^2 u(x)}{\partial x_i \partial x_j} \right)^2 dx$$

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3. Methodology: Classical Method

- **Diffeomorphic Demons¹:**

Gradually deforms the moving image until it matches the fixed image.

- Estimate a small deformation update.
- Smooth the update.
- Compose it with the current transformation.
- Repeat for several iterations.

Diffeomorphic Demons Registration

- 1 Initialize $\phi \leftarrow \text{Id}$
 - 2 Warp: $M \circ \phi$
 - 3 Compute update field u
 - 4 Smooth update: $u \leftarrow K_{\text{fluid}} * u$
 - 5 Compose: $c \leftarrow \phi \circ \exp(u)$
 - 6 Smooth transform: $\phi \leftarrow K_{\text{diff}} * c$
 - 7 Repeat for N iterations
 - 8 Return $M \circ \phi$
-

Algorithm 1. Diffeomorphic Demons pseudo-code.¹

3. Methodology: Metaheuristic Methods - PSO

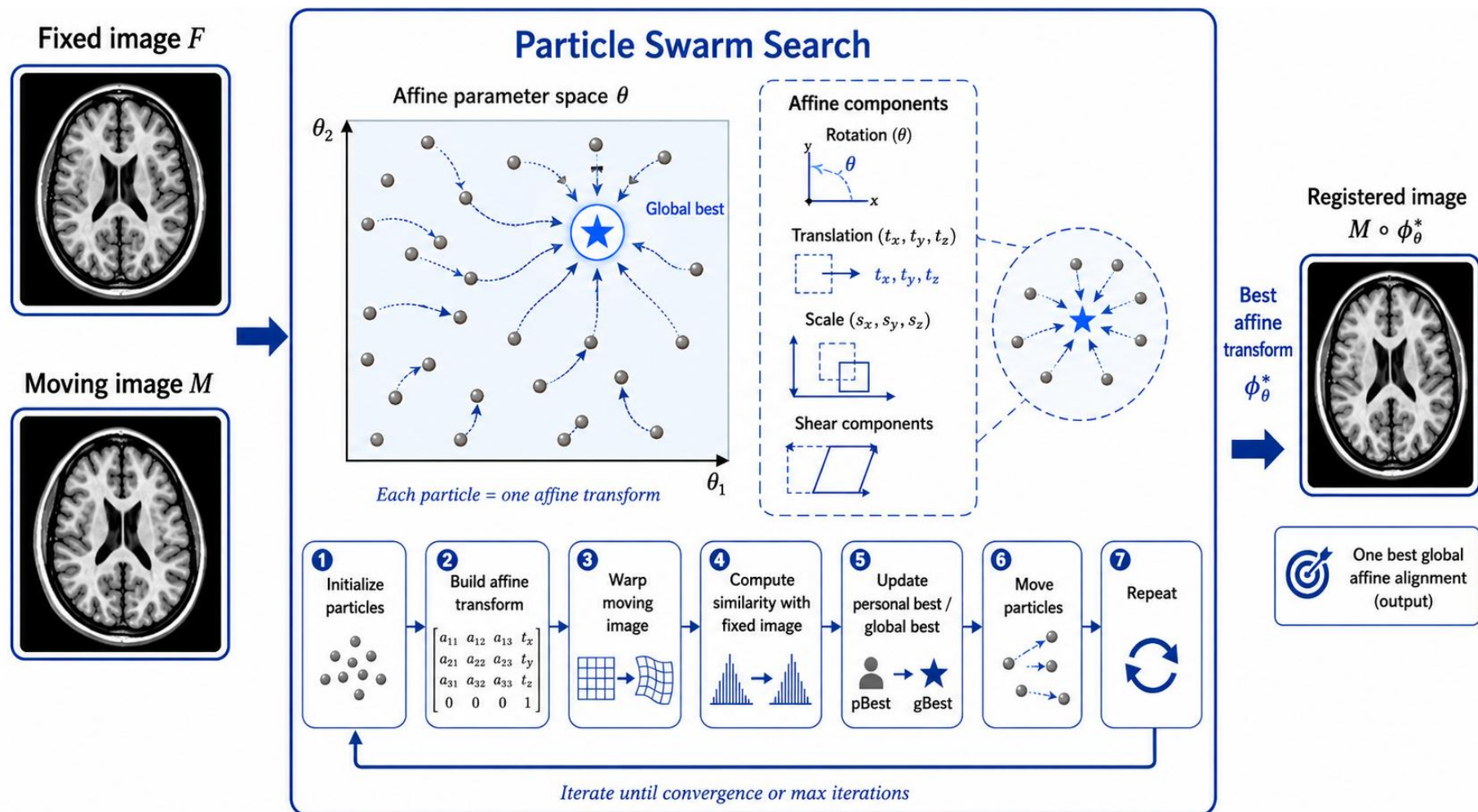


Figure 8. PSO affine registration workflow. Each particle represents a candidate affine transformation, and the swarm iteratively searches for the best transformation that aligns the moving image with the fixed image.¹

3. Methodology: VoxelMorph

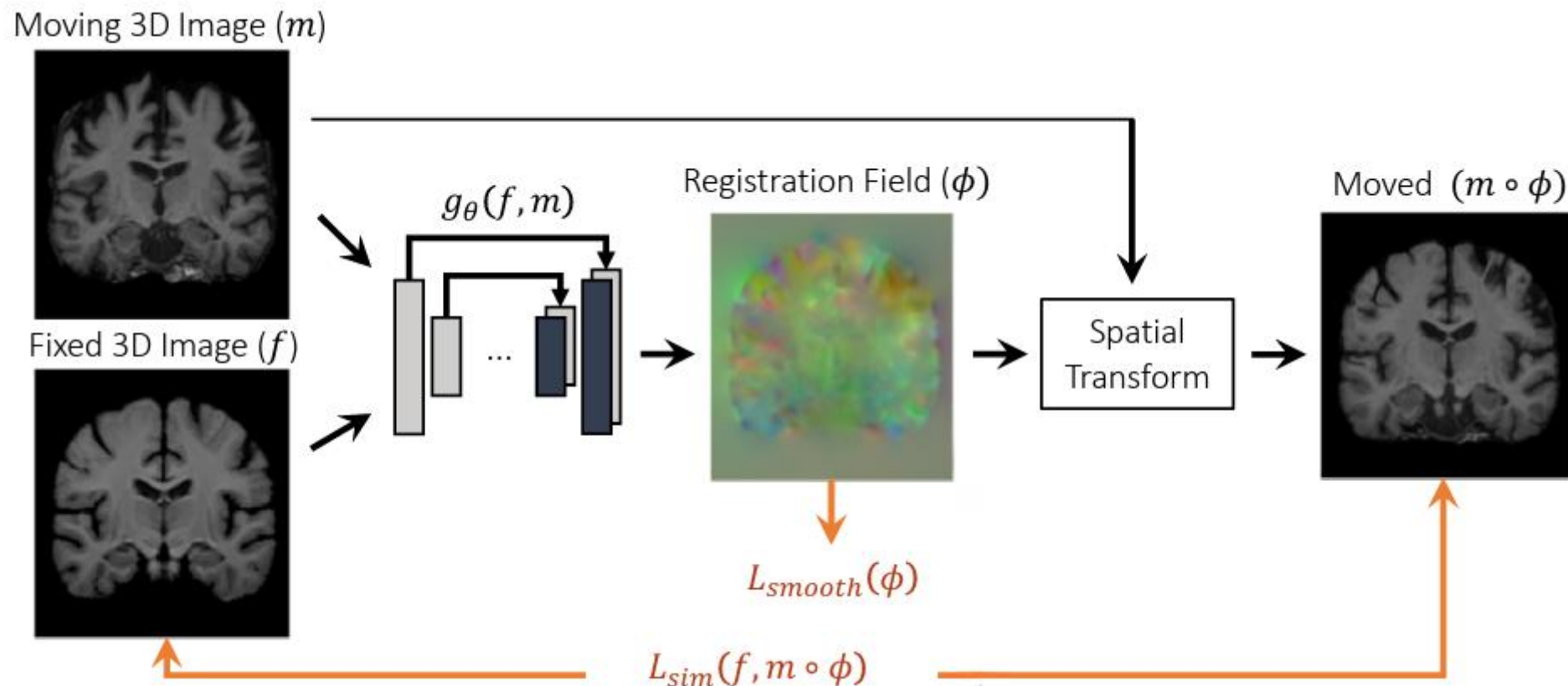


Figure 9. Overview of VoxelMorph. The network learns parameters θ of a function $g_{\theta}(f, m)$, which estimates a registration field ϕ between a moving 3D volume m and a fixed 3D volume f . The moving image is then warped by a spatial transformer to produce the registered image $m \circ \phi$.¹

3. Methodology: TransMorph

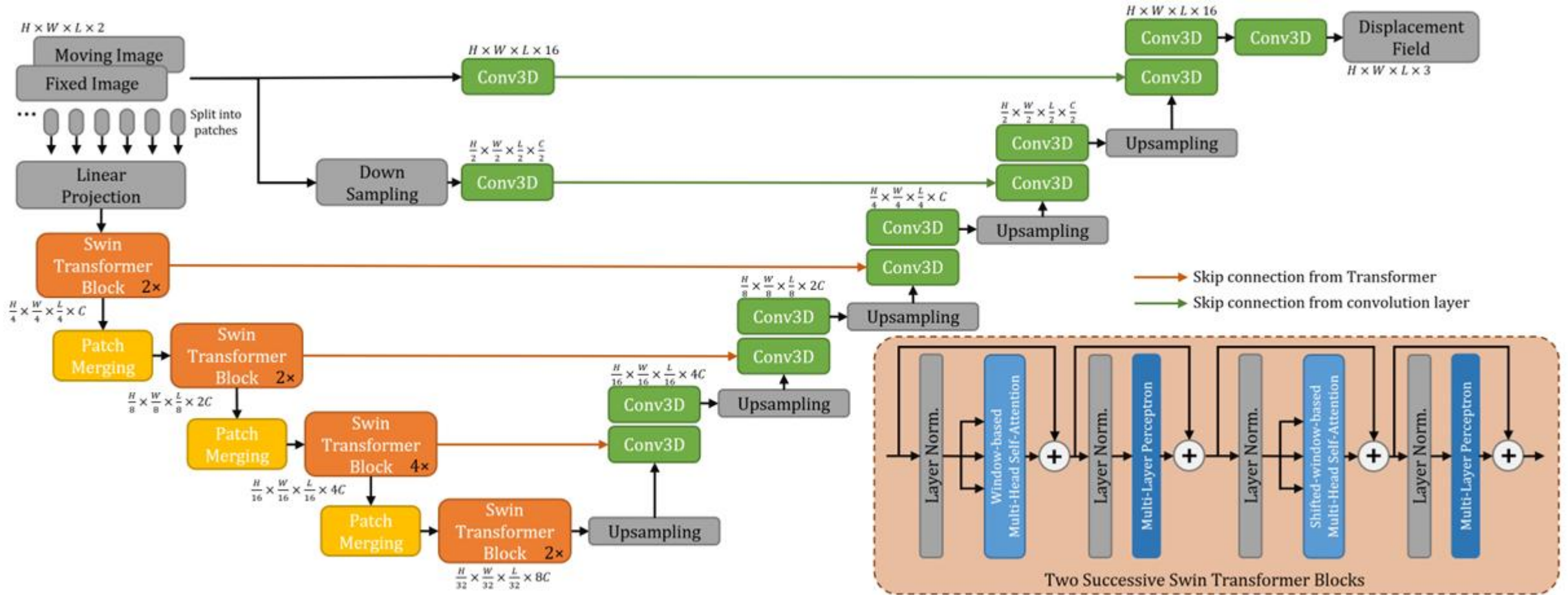


Figure 10. Conceptual TransMorph architecture. Fixed and moving 3D images are encoded into patch-level features, processed by Transformer blocks to capture long-range spatial context, and decoded into a dense displacement field for image registration.¹

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4. Dataset: OASIS¹

- OASIS-1 3D brain MRI volumes
- Public 3D brain MRI dataset
- T1-weighted scans from normal and demented subjects
- 424 adjacent image pairs
- Resolution: $64 \times 64 \times 64$

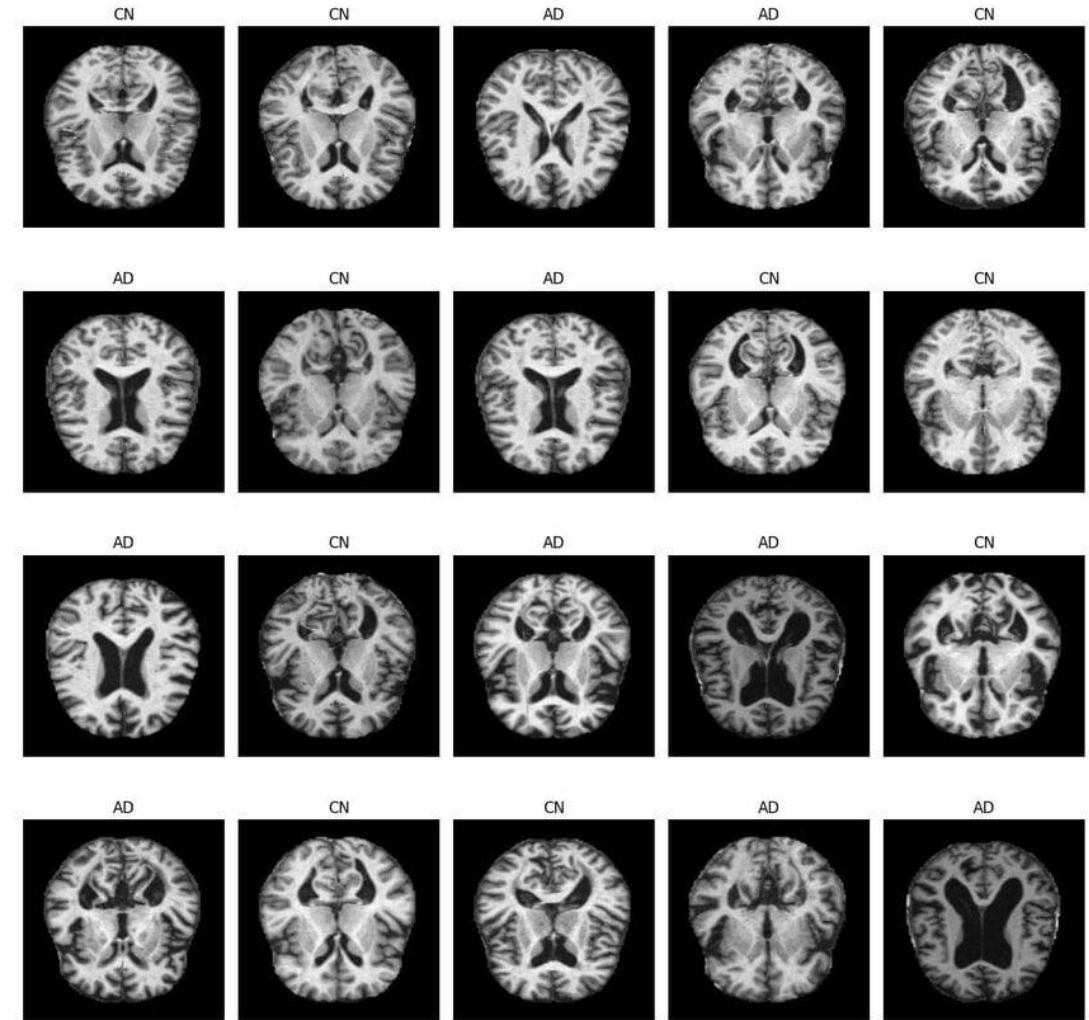


Figure 11. Example samples from the OASIS dataset.

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5. Experiment: Settings

- Hardware:
 - **CPU:** Intel Xeon Gold 6244, 16 cores
 - **RAM:** 64GB
 - **GPU:** Quadro RTX 6000, 24 GB VRAM
- Software:
 - **Python** 3.12.13
 - **PyTorch** 2.10.0 + CUDA 12.6

Table 1. Hyperparameter configuration for 4 methods

Method	Parameter	Value
Classical	Transform	Dense deformation
Classical	Iterations	40
Classical	Smoothing sigma	1.3
PSO	Transform	3D affine
PSO	Particles	16
PSO	Iterations	40
PSO	Optimized parameters	Rotation, translation, scale, shear
PSO	Rotation bound	Approx. $\pm 8^\circ$
PSO	Translation bound	Approx. $\pm 0.08 \times$ image dimension
PSO	Scale bound	0.90 to 1.10
PSO	Shear bound	± 0.08
VoxelMorph	Base channels	4
VoxelMorph	Epochs	20
VoxelMorph	Batch size	2
VoxelMorph	Learning rate	10^{-3}
VoxelMorph	Smoothness weight	0.05
TransMorph	Patch size	8
TransMorph	Embed dim	24
TransMorph	Transformer depth	1
TransMorph	Attention heads	4
TransMorph	Decoder channels	12
TransMorph	Epochs	20
TransMorph	Batch size	2
TransMorph	Learning rate	10^{-3}
TransMorph	Smoothness weight	0.05

5. Experiment: Quantitative Results

- **VoxelMorph** achieves the best intensity-based performance in **MSE** and **NCC**.
- **PSO** obtains the highest **Dice¹** score, but requires the longest runtime.

Table 2. Quantitative comparison of four registration methods on 85 OASIS-1 test pairs.

Method	Sec ↓	MSE ↓	NCC ↑	Dice ↑
Classical	0.512	0.0137 ± 0.0081	0.732 ± 0.131	0.164 ± 0.0369
PSO affine	9.362	0.0109 ± 0.0109	0.793 ± 0.192	0.331 ± 0.2035
VoxelMorph	0.363	0.0089 ± 0.0129	0.827 ± 0.226	0.165 ± 0.0374
TransMorph	0.385	0.0099 ± 0.0119	0.807 ± 0.206	0.219 ± 0.0917

5. Experiment: Summary Metrics

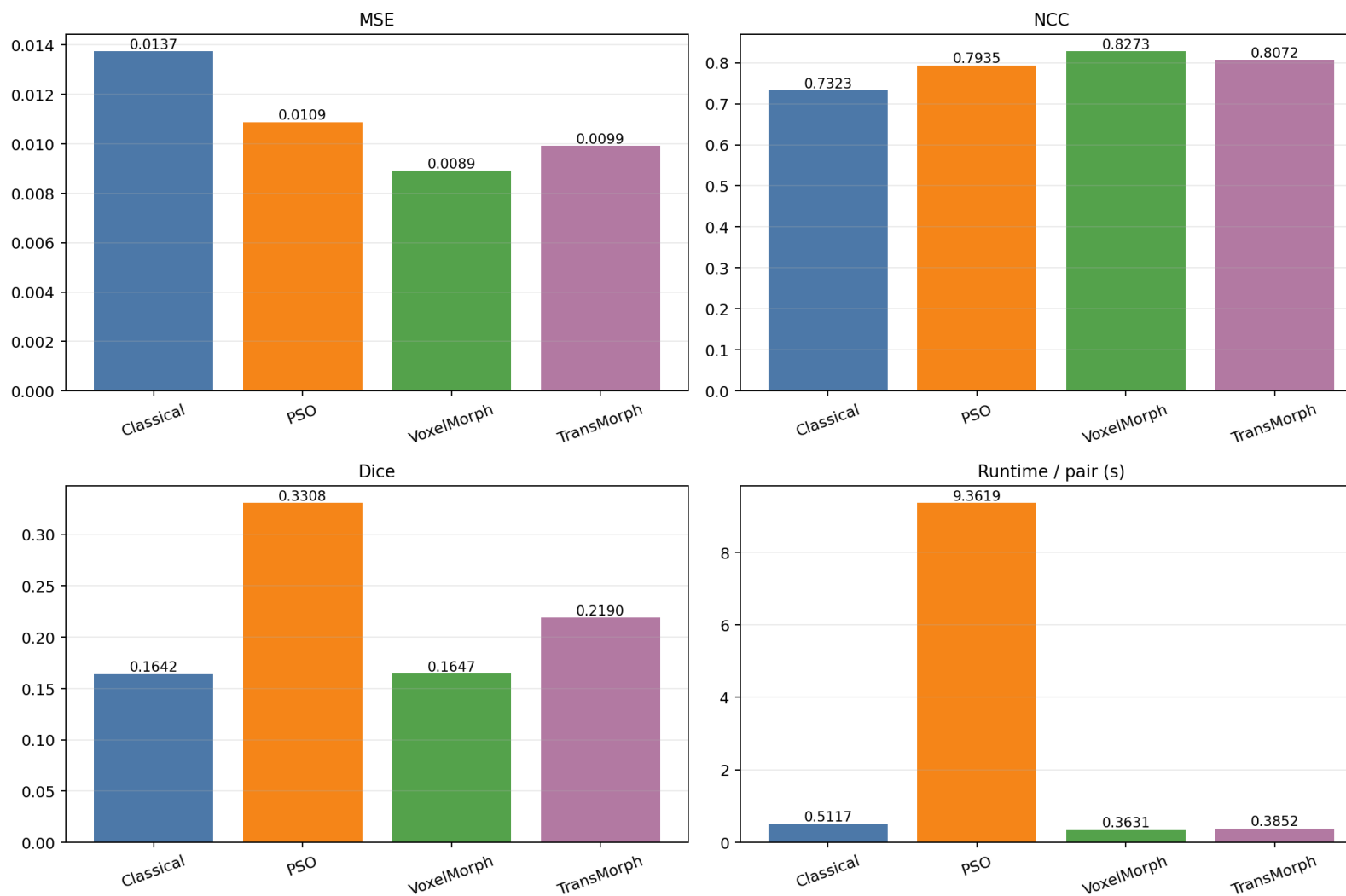


Figure 12. Mean test performance of the methods in terms of MSE, NCC, Dice, and runtime.

5. Experiment: NCC–Dice Comparison

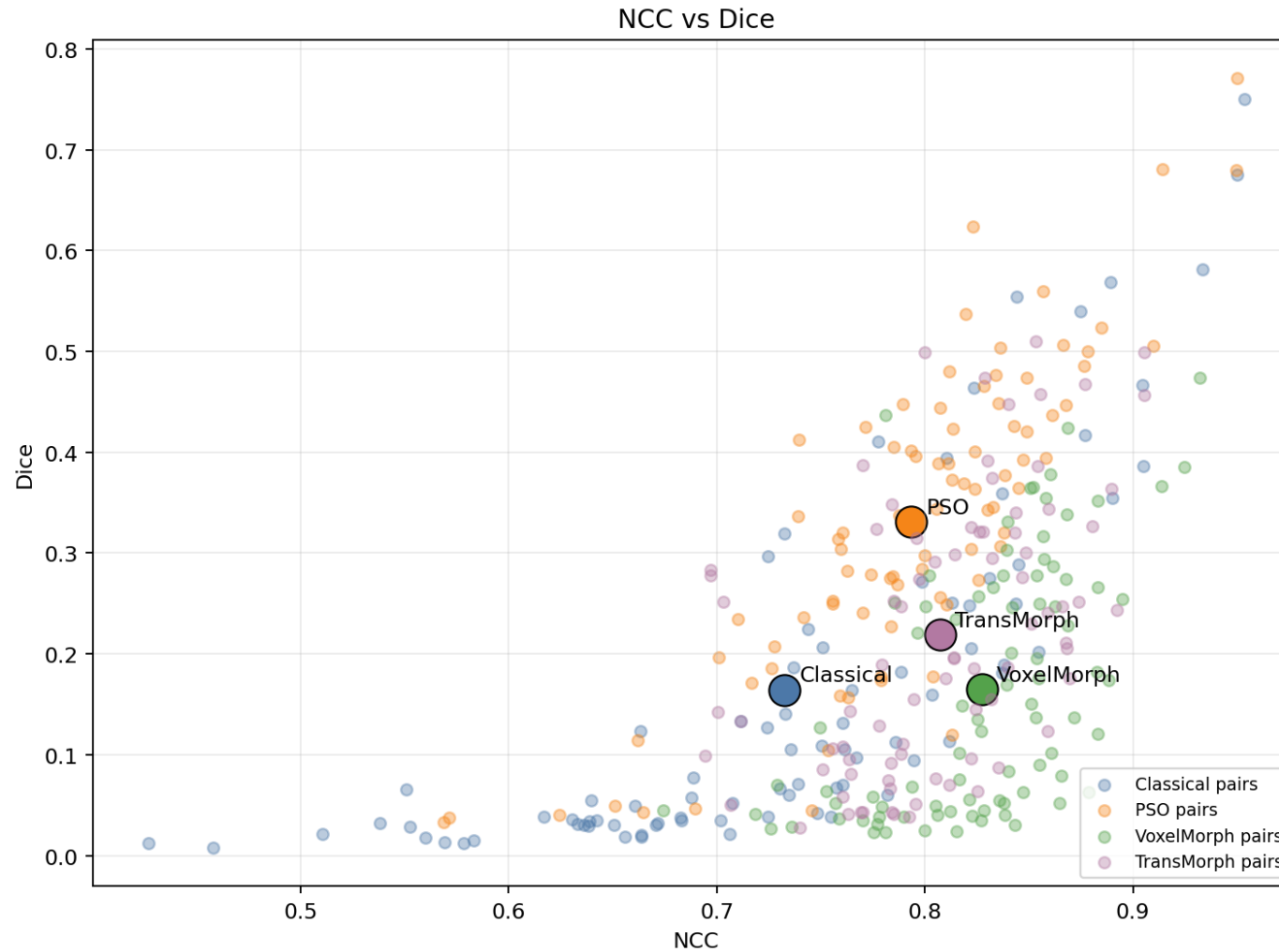


Figure 13. Scatter plot of after-registration NCC versus Dice score for all test pairs.

5. Experiment: Training Loss Curves

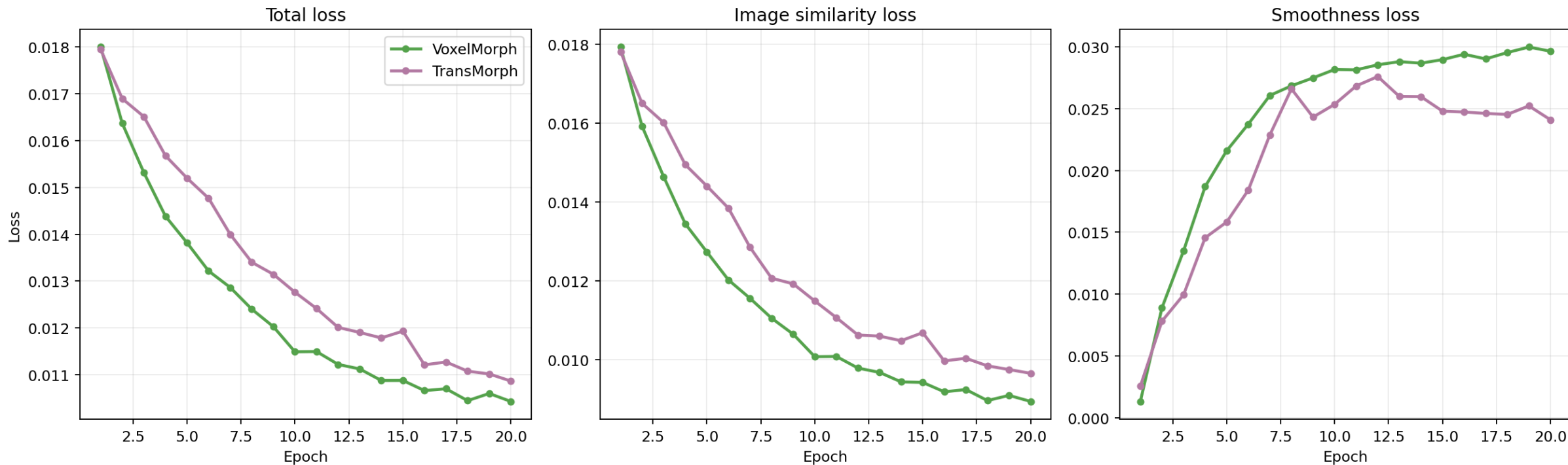


Figure 14. Training loss curves of VoxelMorph and TransMorph over 20 epochs.

5. Experiment: Registration Visualization

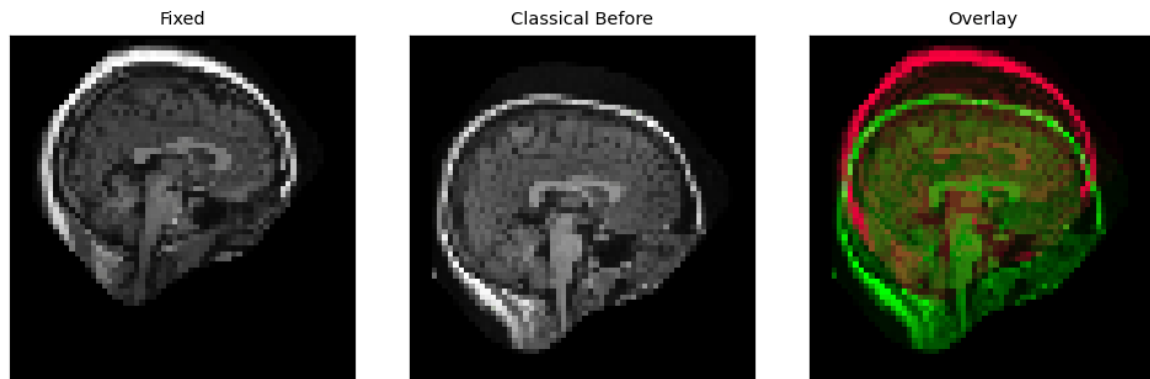


Figure 15. Registration Visualization for Classical Method

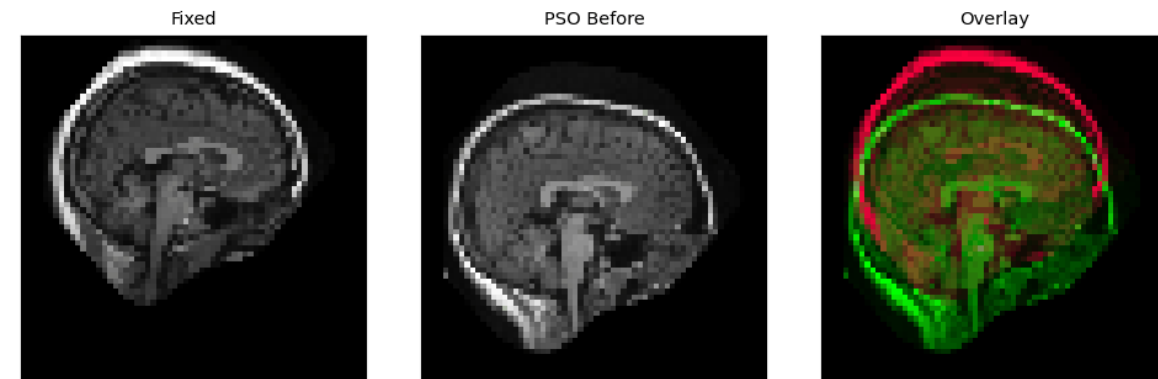


Figure 16. Registration Visualization for PSO

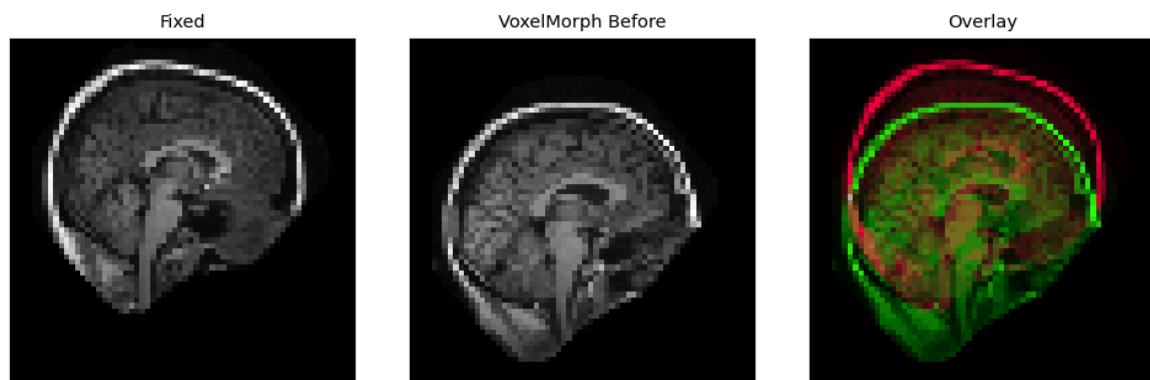


Figure 17. Registration Visualization for VoxelMorph

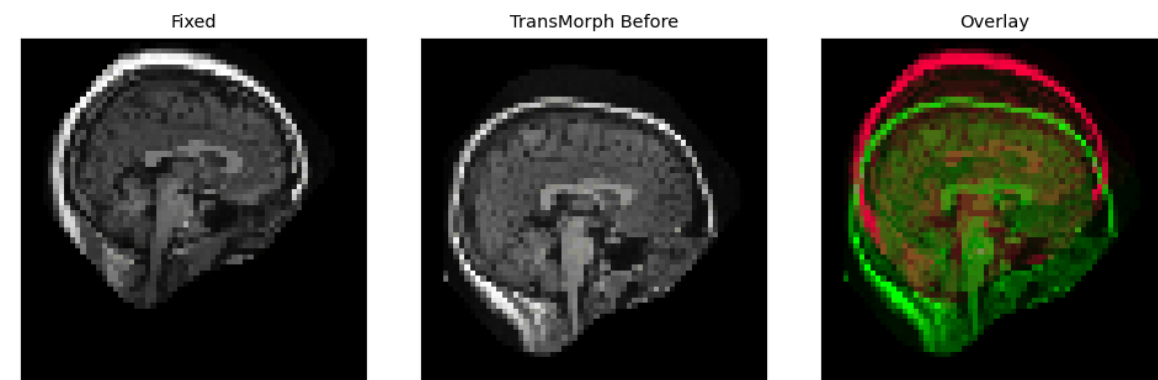


Figure 18. Registration Visualization for TransMorph

5. Experiment: Axial Slice Sweep Visualization

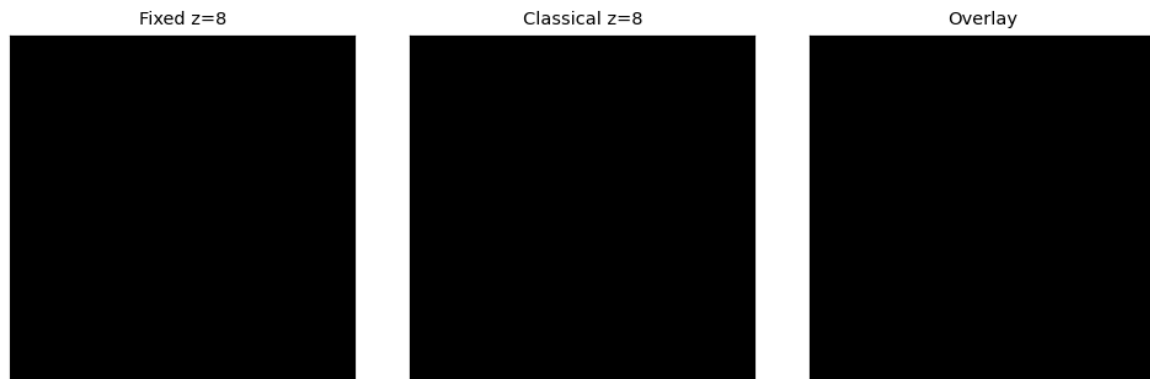


Figure 19. Axial Slice Visualization for Classical Method

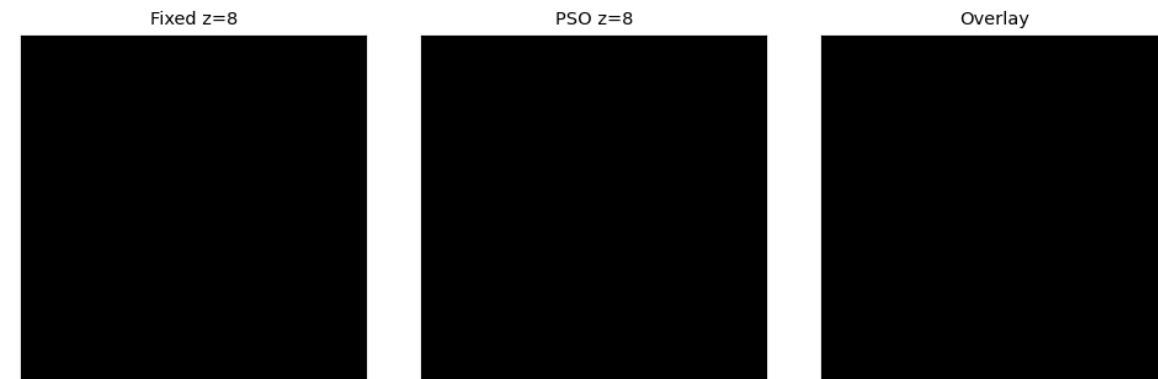


Figure 20. Axial Slice Visualization for PSO

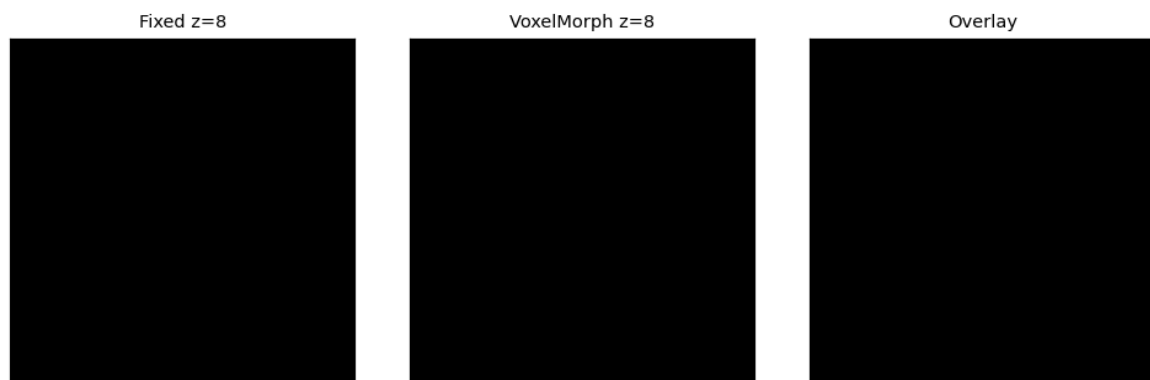


Figure 21. Axial Slice Visualization for VoxelMorph

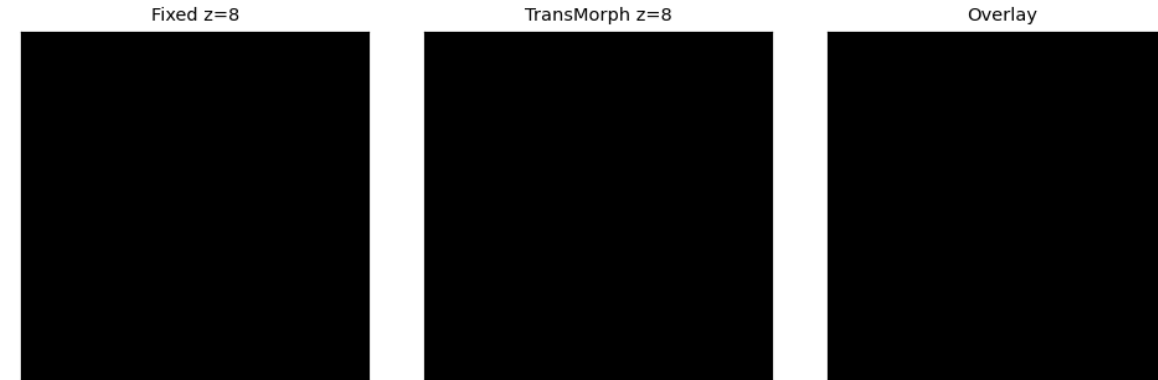


Figure 22. Axial Slice Visualization for TransMorph

5. Experiment: PSO Iteration Progress Visualization

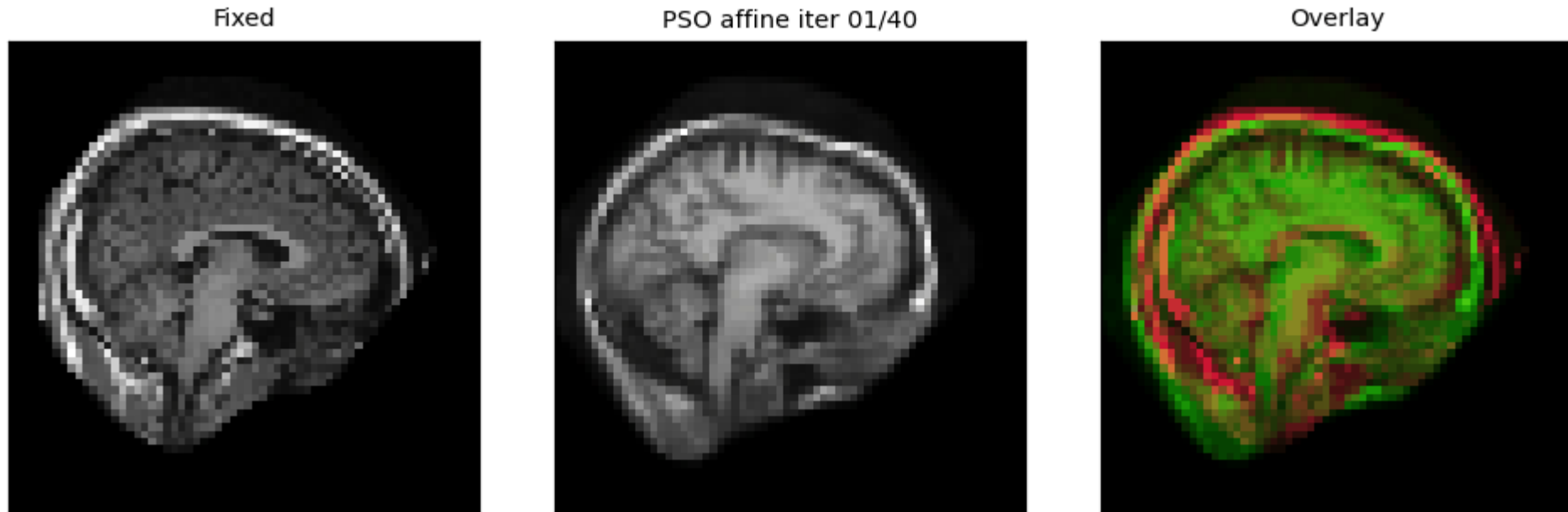


Figure 23. PSO registration progress for rigid registration

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6. Conclusion

- Studied the fundamental problem of Biomedical Image Registration.
- Implemented and compared four representative methods: Diffeomorphic Demons, PSO, VoxelMorph, and TransMorph.
- Conducted experiments on the OASIS-1 3D brain MRI dataset.
- Evaluated performance and developed multiple visualizations

Table 3. Task contributions of each member

Name	Task	Contribution
Pham Cong Hoang	Leader, Slide, Report, Figure visualization, VoxelMorph implementation	25%
Le Tien Hop	Data Processing, Classical Method implementation, Report	25%
Tran Phong Quan	TransMorph implementation, Report, Survey	25%
Luu Hieu An	PSO implementation, Figure visualization	25%

THANK YOU!

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